

The ordering is a bit weird and also I started working on this unit last year so the required problems might have been changed since then

4 hours on examples

Problem 1 (08SGPTST)

For some a, b real numbers, $f(x) \equiv ax^2 + bx$. We will show that this works and is unique in the following proof.

Note that if $f(x)$ is a solution, then so is $g(x) = f(x) + ax^2 + bx$ since,

$$(x+y)(f(x) - f(y)) = (x-y)f(x+y)$$

$$\text{As } (x+y)(ax^2 - ay^2) = (x-y)(a(x+y)^2),$$

$$\Leftrightarrow (x+y)(f(x) + ax^2 - f(y) - ay^2) = (x-y)(f(x+y) + a(x+y)^2)$$

$$\text{As } (x+y)(bx - by) = (x-y)(b(x+y)),$$

$$\begin{aligned} \Leftrightarrow (x+y)(f(x) + ax^2 + bx - f(y) - ay^2 - by) \\ = (x-y)(f(x+y) + a(x+y)^2 + b(x+y)) \end{aligned}$$

$$\Leftrightarrow (x+y)(g(x) - g(y)) = (x-y)g(x+y)$$

Plugging in $(x, -x)$, we get $f(x) = 0$. WLOG shift f such that $f(1) = f(-1) = 1$. We plug in $(x, 1)$ to get,

$$(x+1)(f(x) - f(1)) = (x-1)f(x+1)$$

$$\Leftrightarrow (x+1)f(x) - (x-1)f(x+1) = x+1$$

We plug in $(x+1, -1)$ to get,

$$x(f(x+1) - f(-1)) = (x+2)f(x)$$

$$\Leftrightarrow -(x+2)f(x) + xf(x+1) = x$$

We may solve the system of equations to get $f(x) \equiv x^2$. This clearly satisfies the equation. In general, prior to the shift, $f(x) \equiv ax^2 + bx$ for some a, b and all will work.

Problem 3 (13BALK3)

We claim the answer is $f(x, y, z) \equiv \frac{y + \sqrt{y^2 + 4xz}}{2x}$. We verify that this satisfies each condition.

a) $x \frac{y+\sqrt{y^2+4xz}}{2x} = z \frac{y+\sqrt{y^2+4xz}}{2z}$ is true.

b) $\frac{ky+\sqrt{k^2y^2+4k^2xz}}{2x} = k \left(\frac{y+\sqrt{y^2+4xz}}{2x} \right)$ is true.

c) $\frac{k+\sqrt{k^2+4(k+1)}}{2} = \frac{k+(k+2)}{2} = k+1$ is true.

Setting $k = \frac{y+\sqrt{y^2+4xz}}{2z} \Rightarrow k^2 = \frac{y^2+2xz+\sqrt{y^2+4xz}}{2z^2}$ and using the second equation,

$$f(x, y, z) = \frac{2z}{y + \sqrt{y^2 + 4xz}} f\left(x, \frac{y + \sqrt{y^2 + 4xz}}{2z} y, \frac{y^2 + 2xz + y\sqrt{y^2 + 4xz}}{2z^2} \cdot z^2\right)$$

Claim: $f(x, y, z) = f(mx, my, mz)$ for any positive m .

Proof:

$$\begin{aligned} f(k^2x, k^2y, k^2z) &= \frac{z}{x} f(k^2z, k^2y, k^2x) = \frac{kz}{x} f(k^2z, ky, x) \\ &= \frac{1}{k} f(x, ky, kz^2) = f(x, y, z) \quad \square \end{aligned}$$

Therefore,

$$\begin{aligned} &\frac{2z}{y + \sqrt{y^2 + 4xz}} f\left(x, \frac{y + \sqrt{y^2 + 4xz}}{2z} y, \frac{y^2 + 2xz + y\sqrt{y^2 + 4xz}}{2z^2} \cdot z^2\right) \\ &= \frac{2z}{y + \sqrt{y^2 + 4xz}} f\left(x, \frac{y^2 + y\sqrt{y^2 + 4xz}}{2z}, \frac{y^2 + 2xz + y\sqrt{y^2 + 4xz}}{2z}\right) \\ &= \frac{2z}{y + \sqrt{y^2 + 4xz}} f\left(1, \frac{y^2 + y\sqrt{y^2 + 4xz}}{2xz}, \frac{y^2 + 2xz + y\sqrt{y^2 + 4xz}}{2xz}\right) \end{aligned}$$

Since $\frac{y^2+2xz+y\sqrt{y^2+4xz}}{2xz} = \frac{y^2+y\sqrt{y^2+4xz}}{2xz} + 1$, we use equation 3 to get,

$$\begin{aligned} &= \frac{2z}{y + \sqrt{y^2 + 4xz}} \left(\frac{y^2 + 2xz + y\sqrt{y^2 + 4xz}}{2xz} \right) = \frac{1}{k} \cdot \left(k^2 \cdot \frac{z}{x} \right) \\ &= k \cdot \frac{z}{x} = \frac{y + \sqrt{y^2 + 4xz}}{2x} \end{aligned}$$

4 clubs; 6 hours

Problem 6 (00AMO1)

FTSOC, let there exists such a function.

Claim: Given $f\left(x + \frac{d}{2}\right) \leq \frac{f(x)+f(x+d)}{2} - md \forall$ real numbers $x, d > 0$, we have $f\left(x + \frac{d}{2}\right) \leq \frac{f(x)+f(x+d)}{2} - \frac{3}{2}md$.

Proof: We have,

$$\begin{aligned} f\left(x + \frac{d}{4}\right) &\leq \frac{f(x) + f\left(x + \frac{d}{2}\right)}{2} - \frac{md}{2} \\ &\leq \frac{3f(x) + f(x+d)}{4} - md \end{aligned}$$

Similarly,

$$f\left(x + \frac{3d}{4}\right) \leq \frac{f(x) + 3f(x+d)}{4} - md$$

Therefore, adding the two expressions,

$$f(x) + f(x+d) - 2md \geq f\left(x + \frac{d}{4}\right) + f\left(x + \frac{3d}{4}\right)$$

However, using the initial expression,

$$f\left(x + \frac{d}{4}\right) + f\left(x + \frac{3d}{4}\right) \geq 2\left(f\left(x + \frac{d}{2}\right) + \frac{md}{2}\right)$$

Therefore, as desired,

$$\begin{aligned} f(x) + f(x+d) - 2md &\geq 2\left(f\left(x + \frac{d}{2}\right) + \frac{md}{2}\right) \\ \Leftrightarrow f(x) + f(x+d) - 3md &\geq 2f\left(x + \frac{d}{2}\right) \\ \Leftrightarrow f\left(x + \frac{d}{2}\right) &\geq \frac{f(x) + f(x+d)}{2} - \frac{3md}{2} \quad \square \end{aligned}$$

Plugging in $y = x + d$,

$$f\left(x + \frac{d}{2}\right) \leq \frac{f(x) + f(x+d)}{2} - d$$

Repeatedly applying the claim, for any integer k and real numbers $x, d > 0$.

$$f\left(x + \frac{d}{2}\right) \leq \frac{f(x) + f(x+d)}{2} - \left(\frac{3}{2}\right)^k d$$

For any finite value of $f\left(x + \frac{d}{2}\right)$, we may take a sufficiently large k to get a contradiction.

Problem 7 (09IMO5)

Claim: $f(1) = 1$

Proof: FTSOC, if $f(1) = k > 1$, plug in $a = 1$ to get $1, f(b), f(b+k-1)$ form a non-degenerate triangle. We must have $f(b) = f(b+k-1)$ for all b . Therefore, we fix the maximum $m = \max(f(1), f(2), \dots, f(k-1))$. Note that $f(x) \leq m$ for all x .

We set $a = 1434m$ to see that $1434m, f(b), f(b+f(1434m)-1)$ form a triangle implying $2m \geq f(b) + f(b+f(1434m)-1) > 1434m$. (!) $\square$

Using the claim, we plug in $b = 1$ to get $a, 1, f(f(a))$ form a triangle. Therefore, for all x ,

$$f(f(a)) = a$$

Note that f is bijective. Let $f(k) = 2$ for some $k > 1$. Plugging in $(f(a), k)$, we have $f(a), 2, f(a+k-1)$. Therefore, $|f(a+k-1) - f(a)| = 1$. As a conclusion, for any $1 \leq n \leq k-1$ and positive m ,

$$f(n + m(k-1)) \leq f(n) + m$$

We fix $f(1), \dots, f(k-1)$ and look at the upperbound of $\frac{f(x)}{x}$ as x goes to infinity. For some $1 \leq n \leq k-1$ such that $n \equiv x \pmod{k-1}$,

$$\frac{f(x)}{x} \leq \frac{f(n) + \left\lfloor \frac{x}{k-1} \right\rfloor - 1}{x}$$

The limit of the RHS approaches $\frac{1}{k-1} \leq 1$ as x goes to infinity. Additionally, the RHS is always less than $f(n)$ so $f(x) \leq x \max(f(1), f(2), \dots, f(k-1))$ also.

Lemma 1: Given $f(a) \leq a-1$, $f(2a-1) \leq 2a-3$.

Proof: Let $f(a) = b \leq a-1$. Note that $f(b) = a$ for $f(f(a)) = a$. Plugging (b, a) , we know $b, b, f(2a-1)$ form a triangle,

$$b + b \geq f(2a-1) - 1$$

$$\Leftrightarrow 2b - 1 \geq f(2a-1)$$

$$\Rightarrow 2a - 3 \geq f(2a - 1) \quad \square$$

We now prove that $f(x) \equiv x$. For the sake of contradiction, let this not be the case meaning $f(a) = b, f(b) = a$ for some $b < a$. Noting that $f(a) \leq a - 1$, we repeatedly apply the lemma k times to get $f(2^k(a - 1) + 1) = x \leq 2^k(a - 2) + 1$. We have $x \leq \frac{a-2}{a-1}(2^k(a - 1) + 1)$. Therefore, there exists arbitrarily large d such that,

$$f(d) \leq \frac{a-2}{a-1}d$$

If we let $f(d) = x$, we have previously proved that $f(x) = d \leq x \max(f(1), f(2), \dots, f(k - 1))$ so $x \geq \frac{d}{\max(f(1), f(2), \dots, f(k-1))}$. As d gets arbitrarily large, x will also get arbitrarily large. We know $x \leq \frac{a-2}{a-1}d \Leftrightarrow f(x) \geq \frac{a-1}{a-2}x \Leftrightarrow \frac{f(x)}{x} \geq \frac{a-1}{a-2}$ which is impossible since we've proved the bound of $\frac{f(x)}{x}$ approaches 1.

10 clubs; 9 hours

Problem 8 (10BWAY)

We claim the solutions are $f(x) \equiv 0$, $f(x) \equiv -x$, and $f(x) \equiv 2 - x$. We can check that all three works.

We prove no other solutions exist. We begin by plugging in $(0, 0)$ to get,

$$f(0) + f(0) = f(0)^2 \Leftrightarrow f(0) \in \{0, 2\}$$

Case 1: $f(0) = 2$

Plug in $(0, x)$ to get,

$$f(0) + f(0) = f(0)f(x) + xf(0)$$

$$\Leftrightarrow 4 = 2f(x) + 2x$$

$$\Leftrightarrow f(x) = 2 - x$$

This is indeed one of the solutions.

Case 2: $f(0) = 0$

We plug in $(x, 0)$ to get,

$$f(x^2) = xf(x)$$

Plugging in $(-x, 0)$, we see $f(x^2) = -xf(-x)$ also so $f(-x) = -f(x)$ and f is odd.

Now plug in $(x, -x)$ to see,

$$f(x^2) + f(-x^2) = f(x)f(-x) - xf(x) + xf(0)$$

$$\Leftrightarrow 0 = -f(x)^2 - xf(x)$$

$$f(x) \in 0, -x$$

We resolve the point-wise trap to finish. Assume for the contradiction, there exists some $f(a) = 0$ but $f(b) = -b$ for non-zero real numbers a, b . Plug in $x = a$ into $f(x^2) = xf(x)$ to get $f(a^2) = 0$. We now plug in $(a, \frac{b}{a})$ into the original equation to get,

$$f(a^2) + f(b) = f(a)f(y) + yf(a) + af\left(a + \frac{b}{a}\right)$$

$$\Leftrightarrow f(b) = af\left(a + \frac{b}{a}\right)$$

$$\Leftrightarrow -\frac{b}{a} = f\left(a + \frac{b}{a}\right)$$

We know that $-\frac{b}{a} \neq 0$ so,

$$-\frac{b}{a} = -a - \frac{b}{a}$$

$$\Leftrightarrow a = 0 (!)$$

Problem 9 (19TSTST1)

We claim the solutions are $a \diamond b = \frac{a}{b}$ or $a \diamond b = ab$. Both clearly work.

We define $f_a(b) = a \diamond b$ and $g_a(b) = b \diamond a$.

Lemma 1: $f_a(b)$ is bijective for all a .

Proof: Looking at $a \diamond (b \diamond c) = (a \diamond b) \cdot c$, we can prove injective since $(b \diamond c) = (b \diamond c')$ would imply,

$$(a \diamond b) \cdot c = a \diamond (b \diamond c) = a \diamond (b \diamond c') = (a \diamond b) \cdot c'$$

Since $(a \diamond b) > 0$, this implies $c = c'$. We can prove surjective since $a \diamond (b \diamond c) = (a \diamond b) \cdot c$ gives $(a \diamond b) \cdot c$ is in the range of f_a as it is equal to $f_a(b \diamond c)$. Since $a \diamond b$ is positive, $(a \diamond b) \cdot c$ takes all real numbers as desired. $\square$

Lemma 2: $g_a(b)$ is bijective for all a .

Proof: Note that,

$$a \diamond ((b \diamond c) \cdot d) = a \diamond (b \diamond (c \diamond d)) = (a \diamond b) \cdot (c \diamond d)$$

For injectivity, assume that $(b \diamond c) = (b' \diamond c)$. We plug in to get the following,

$$\begin{aligned} (a \diamond b') \cdot (c \diamond d) &= a \diamond ((b' \diamond c) \cdot d) \\ &= a \diamond ((b \diamond c) \cdot d) = (a \diamond b) \cdot (c \diamond d); \end{aligned}$$

Given $(a \diamond b') = (a \diamond b)$, we have $b = b'$ as desired using lemma 1. For surjectivity, we fix a, b, d and vary c in the equation,

$$a \diamond ((b \diamond c) \cdot d) = (a \diamond b) \cdot (c \diamond d)$$

Across values of positive reals c , using lemma 1, $(b \diamond c)$ takes all positive reals so $(b \diamond c) \cdot d$ takes all positive reals meaning $a \diamond ((b \diamond c) \cdot d)$ takes all positive reals.

Given that $(a \diamond b) \cdot (c \diamond d)$ takes all positive reals and since $(a \diamond b)$ is positive, $(c \diamond d)$ takes all positive real values as desired.

Using the claim, we see that,

$$\begin{aligned} a \diamond ((b \diamond c) \cdot d) &= (a \diamond b) \cdot (c \diamond d) \\ a \diamond ((b \diamond c) \cdot kd) &= (a \diamond b) \cdot (c \diamond kd) \end{aligned}$$

Dividing the two equations,

$$\frac{a \diamond ((b \diamond c) \cdot d)}{a \diamond ((b \diamond c) \cdot kd)} = \frac{(c \diamond d)}{(c \diamond kd)}$$

If we fix c, d and let $d(b \diamond c) = x$, noting that x can take any positive real value as b varies, we have for all $a, x, c, k, d > 0$,

$$\frac{a \diamond x}{a \diamond kx} = \frac{c \diamond d}{c \diamond kd}$$

If we plug in $c = a$,

$$\frac{f_a(x)}{f_a(kx)} = \frac{f_a(d)}{f_a(kd)}$$

Note that the equation is homogenous. We let $m \cdot h(x) = f_a(x)$ such that $h(1) = 1$. We plug in $d = 1$ to get,

$$h(kx) = h(x)h(k)$$

We may plug in $h(x) = e^{h'(\log(x))}$ for some function h' taking real numbers to real numbers to get,

$$e^{h'(\log(kx))} = e^{h'(\log(x)) + h'(\log(k))}$$

Substituting $K = \log(k)$; $X = \log(x)$ for some real numbers K, X ,

$$e^{h'(K+X)} = e^{h'(K) + h'(X)}$$

$$\Leftrightarrow h'(K + X) = h'(K) + h'(X)$$

Claim: $f_a(b) \geq \frac{a}{2}$ for all $1 \leq b \leq 2$.

Proof:

$$a \diamond (a \diamond 1) = (a \diamond a) \cdot 1$$

From f_a being injective,

$$\Leftrightarrow a \diamond 1 = a$$

We know that,

$$1 \diamond (1 \diamond a) = (1 \diamond 1) \cdot a = a$$

Therefore,

$$a \diamond (1 \diamond (1 \diamond d)) = (a \diamond 1) \cdot (1 \diamond d)$$

$$\Leftrightarrow a \diamond d = a(1 \diamond d)$$

In the case that $d = a \geq 1$, we have $a \diamond a = a(1 \diamond a) \Leftrightarrow (1 \diamond a) \geq \frac{1}{a}$. Therefore,

$$a \diamond d \geq \frac{a}{d}$$

This gives the claim. $\square$

Since $f_a(x)$ was bounded above or below 1 for $x \geq a$, $h(x)$ was bounded above or below $\frac{1}{m}$ for $x \geq a$, so $h'(x)$ is also bounded above or below for $x \geq a$. By Cauchy's functional equation, $h'(x) = ax$ for some real a . Plugging this in, we'd get $f_a(x) = u \cdot x^v$ for some positive real u and some real v . We now plug $b = a$ to get, for all a, c ,

$$a \diamond (a \diamond c) = (a \diamond a) \cdot c$$

$$\Leftrightarrow a \diamond (u \cdot c^v) = u \cdot a^v \cdot c$$

$$\Leftrightarrow u \cdot (u^v \cdot c^{v^2}) = u \cdot a^v \cdot c$$

Since c is any positive real number, we must have $v^2 = 1 \Leftrightarrow v = \pm 1$.

$$\Leftrightarrow u^v = a^v$$

Since v is non-zero, $u = a$. In conclusion, $a \diamond x \equiv \frac{a}{x}$ or ax as desired for all a, x . There is no point-wise trap since we used Cauchy's functional equations.

18 clubs; 12 hours

Problem 10 (18CHNTST14)

Note that the range of f is exactly the range of g since for any $f(t) = s$ we plug in $y = -g(x) + t$ to get $g(\text{something}) = f(t) = s$ and vice versa. In particular, g is also bounded. The conditions are now symmetric in f and g .

Given $f(m) = f(n)$, we have,

$$f(g(x) + m) = g(f(m) + x) = g(f(n) + x) = f(g(x) + n)$$

Therefore, for any s in the range of f and g , $f(m + s) = f(n + s)$ also. (i)

If $s = 0$ is the only thing in the range of g , then $g(x) \equiv 0$ is clearly periodic. Otherwise, we fix some $s \neq 0$ such that $f(s) = s$.

Claim: For each i , there exists some d such that $f(sa + i) = f(s(a + d) + i)$ for all a .

Proof: For arbitrarily negative m , notice that $f(ms + i), f((m + 1)s + i), f((m + 2)s + i), \dots$ must repeat $f(ms + i) = f((m + k)s + i)$ for some k since the range of f is finite. Using (i), we have $f((m + 1)s + i) = f((m + k + 1)s + i)$ we may induct to see $f(sa + i) = f(s(a + d) + i)$ for all a going forward. $\square$

Taking $i = 0, 1, \dots, s - 1$ and looking at the common differences $d_0, \dots, d_{s-1}$ we may take $d = \text{lcm}(d_i)$ to get $f(x) = f(x + d)$ for all x as desired.

Problem 11 (11SLA3)

We either have $f(x) \equiv x^2 + c$; $g(x) = x$ or $f \equiv g \equiv 0$. Both clearly work.

Plug in $(-x, 2x)$ to get,

$$g(f(x)) = f(-x)$$

Using this,

$$f(-x - y) = f(x) + (2x + y) g(y)$$

Applying $y = 0$,

$$f(-x) = f(x) + 2x g(0)$$

On the other hand, plugging in $(x, -x)$, we have,

$$g(f(0)) = f(x) + x g(-x)$$

$$\Leftrightarrow f(0) - x g(-x) = f(x)$$

Thus,

$$f(-x - y) = f(x) + (2x + y) g(y)$$

$$\Leftrightarrow f(0) + (x + y) g(x + y) = f(0) - x g(-x) + (2x + y) g(y)$$

$$\Leftrightarrow (x + y) g(x + y) = (2x + y) g(y) - x g(-x)$$

Note that plugging in $g(x) = g'(x) + ax + b$ still gives the same thing. Therefore, we may WLOG assume $g(1) = g(-1) = 0$.

We may plug in $(1, x)$ to get,

$$(x + 1) g(x + 1) = (x + 2) g(x)$$

Plug in $(x + 1, -1)$ to get,

$$x g(x) = (4x + 1) g(x + 1)$$

We may solve the system of equations to get $g(x) = 0$. In general, g is linear. Plugging back in, we may get the desired options.

26 clubs; 16 hours

Problem 14 (07BALK2)

We claim the solutions are $f(x) \equiv x + c$ for some c or $f(x) \equiv 0$. Both clearly work.

Note that,

$$f(f(x) + y) - f(f(x) - y) = 4f(x)y$$

Therefore, if f is non-zero, we may plug in x such that $f(x) \neq 0$. Changing y , we see that we may always find a pair (a, b) such that $f(a) - f(b) = k$ for any k .

For this pair, we plug in $x = b; y = f(a) - k$ to get,

$$f(f(b) + f(a) - k) = f(f(b) - f(a) + k) + 4 f(b) (f(a) - k)$$

We then plug in $x = a; y = f(b) - k$ to get,

$$f(f(a) + f(b) - k) = f(f(a) - f(b) + k) + 4 f(a) (f(b) - k)$$

Subtracting the two equations, for any k ,

$$0 = f(0) - f(2k) - 4k f(b) + 4k f(a)$$

$$\Leftrightarrow f(2k) = 4k^2 + f(0)$$

Equivalently, $f(x) \equiv x^2 + c$ as desired.

Problem 15 (23AMO2)

Claim: f is strictly increasing.

Proof: We first check that f is injective. If $f(a) = f(b)$, plug (a, b) and (b, a) to get,

$$f(ab + f(a)) = af(b) + 2$$

$$f(ab + f(b)) = bf(a) + 2$$

Clearly, since f is positive, $a = b$ from $f(a) = f(b)$. We next FTSOC assume $f(a) > f(b)$ but $a < b$. We may plug in $\left(a, \frac{f(a)-f(b)}{b-a}\right)$ and $\left(b, \frac{f(a)-f(b)}{b-a}\right)$ to get,

$$a f\left(\frac{f(a) - f(b)}{b - a}\right) + 2 = b f\left(\frac{f(a) - f(b)}{b - a}\right) + 2$$

This is clearly impossible. $\square$

Let $k = f(1)$ and plug in $x = 1$ to get $f(y + k) = f(y) + 2$. We have,

$$f(1) = k; f(k + 1) = k + 2; f(2k + 1) = k + 4 \dots$$

Using the strictly increasing condition, we know $\left|f(y) - k - \frac{2}{k}(y - 1)\right| < 2$ for all $y \geq 1$. We let $y = 1$ to get,

$$f(x + f(x)) = xk + 2$$

$$\Rightarrow \left| xk + 2 - k - \frac{2}{k}(f(x) + x - 1) \right| < 2$$

$$\Rightarrow \left| xk + 2 - k - \frac{2}{k}\left(k + \frac{2}{k}(x - 1) + x - 1\right) \right| < 2 + \frac{4}{k}$$

$$\Leftrightarrow \left| x \left(k - \frac{4}{k^2} - \frac{2}{k} \right) - k + \frac{4}{k^2} + \frac{2}{k} \right| < 2 + \frac{4}{k}$$

Taking sufficiently large x , we see that if $k - \frac{4}{k^2} - \frac{2}{k} \neq 0$, we'll get a contradiction. Therefore, $k - \frac{4}{k^2} - \frac{2}{k} = 0$. Since k is positive, we may check that $k = 2$ is the only solution. We then have $|f(x) - x - 1| < 2$.

We claim that $f(x) \equiv x + 1$. FTSOC, let there exist some y such that $f(y) = y + 1 + d$ for some $0 < |d| < 2$. Plug this y into the original equation to get,

$$\begin{aligned} f(xy + f(x)) &= xy + x + dx + 2 \\ \Rightarrow |xy + x + dx + 2 - xy - f(x) - 1| &\leq 2 \\ \Rightarrow |xy + x + dx + 2 - xy - x - 1 - 1| &\leq 4 \\ \Leftrightarrow |dx| &\leq 4 \end{aligned}$$

Taking sufficiently large x , we have a contradiction.

34 clubs; 20 hours

Ok... these problems are getting hard... Let's backtrack

Problem 2 (15APMO2)

Let $g(x) = f(2^x)$ for any integer $x \geq 1$. Plugging in $a = 2^x$ and $b = 2^y$ for distinct x, y , we have, for all x, y ,

$$g(x)g(y) = g(2x + 2y)$$

Let $g(1) = a; g(2) = b; g(3) = c$. Note that, for all $k \geq 3$,

$$\begin{aligned} g(1)g(k+1) &= g(2)g(k) = g(2k+4) \\ \Rightarrow g(k+1) &= \frac{g(2)g(k)}{g(1)} = \frac{b}{a}(g(k)) \end{aligned}$$

Therefore, $g(x+3) = c \cdot \left(\frac{b}{a}\right)^x$. If $b \neq a$, we may take sufficiently large x to get g is no longer an integer. Therefore, $b = a$. As a result,

$$c = g(3) = g(4) = g(5) = \dots$$

Since $g(14) = g(3)g(4), c^2 = c$. This is clearly impossible for $c \geq 2$.

Problem 4 (19JMO2)

We show that the answer is precisely $a = \pm b$.

For attainability, let $f(x) = x + a$; $g(x) = x$ if $a = b$. Let $f(x) = -x + a$ and $g(x) = -x$ if $a = -b$.

Note that f, g are clearly bijective. FTSOC, let $a \neq \pm b$. WLOG, let $|b| > |a|$.

$$f(g(f(x))) = f(x + b) = f(x) + a$$

Therefore, for all x ,

$$f(x + b) \equiv f(x) \pmod{a}$$

Out of $f(0), f(1), \dots, f(b - 1)$, by Pigeonhole, some leave the same remainder modulo a . From $f(i) \equiv f(j) \pmod{a}$, we have some $f(i + kb) = f(j)$. Since $i \not\equiv j \pmod{b}$, $i \neq j$ and we have injective condition violated.

We now show that a, b are even.

38 clubs; 21 hours

Problem 5 (19IDOTSJ1)

Note that,

$$\begin{aligned} f^{f(2)}(2) &= 4 \\ \Leftrightarrow f(f^{f(2)-1}(2)) &= 4 \end{aligned}$$

Let $k_4 = f^{f(2)-1}(2)$ and define $k_9 = f^{f(2)-1}(3)$ noting that $f(k_9) = 9$. We know,

$$f^{(4)}(m) = f^{f(k_4)}(m) = m^{k_4}; f^{(9)}(m) = f^{f(k_9)}(m) = m^{k_9}$$

Note that,

$$\begin{aligned} f^{(4)}(f^{(4)}(m)) &= m^{k_4^2} \\ f^{(12)}(m) &= f^{(4)}(f^{(8)}(m)) = m^{k_4^3} \end{aligned}$$

Inducting, $f^{(4j)}(m) = m^{k_4^j}$. Also,

$$f^{(4)}(f^{(4)}(f(m))) = f^{(9)}(m)$$

$$\Leftrightarrow f(m)^{k_4^2} = m^{k_9}$$

Since $f(m), m > 1$,

$$\Leftrightarrow f(m) = m^{\frac{k_9}{k_4^2}}$$

Clearly, if we plug in $m = 2$, we see that $\frac{k_9}{k_4^2}$ is a fixed integer k .

$$f(m) = m^k$$

We may plug in directly to see no solutions.

Problem 20 (20USEMO4)

Claim 1: $f(y) \geq y$.

Proof: FTSOC, if $f(y) < y$, we may plug in $x = 1 - \frac{f(y)}{y}$. Since,

$$\begin{aligned} x + f(y) + xy &= x + y \\ \Leftrightarrow \left(1 - \frac{f(y)}{y}\right) + f(y) + \left(1 - \frac{f(y)}{y}\right)y &= 1 - \frac{f(y)}{y} + y \\ \Leftrightarrow f(y) + y - f(y) &= y \end{aligned}$$

This is true, so $f(x + f(y) + xy) = f(x + y)$ as well meaning $xf(y) = 0$. (!)

Using the claim, we may now define $f(x) = g(x) + x$, noting that g goes $R^+ \rightarrow R^{\geq 0}$. We have,

$$\begin{aligned} f(x + f(y) + xy) &= xf(y) + f(x + y) \\ \Leftrightarrow g(g(y) + xy + x + y) &= (x - 1)g(y) + g(x + y) \end{aligned}$$

We plug in (y, x) into the above equation to see,

$$g(g(x) + xy + x + y) = (y - 1)g(x) + g(x + y)$$

Therefore, if $g(x) = g(y)$, we know $(x - 1)g(y) = (y - 1)g(x)$ meaning either $g(x) = g(y) = 0$ or $x = y$.

We also plug in $x = 1$ to get,

$$g(g(y) + 2y + 1) = g(1 + y)$$

Therefore, either $g(y) + 2y + 1 = 1 + y$, which is impossible, or $g(1 + y) = g(g(y) + 2y + 1) = 0$. As a result, $g(y) = 0$ for all $y > 1$.

We now plug in $x > 1$ and $y < 1$ to get, $g(g(x) + xy + x + y) = 0$; $g(x + y) = 0$ meaning $g(y) = 0$ as well. As desired, $g \equiv 0$.

45 clubs; 24 hours

I'll be back for AK o7